

Rama Krishna Molakaseema Data Engineer

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Summary

Results-driven Data Engineer with 3+ years of experience building scalable, cloud-native data solutions across AWS and Azure. Specialized in real-time stream processing, big data pipelines, and end-to-end ETL workflows using Python, SQL, Apache Spark, and Kafka. Adept at integrating diverse data sources, automating processes, and delivering actionable insights through Power BI, Tableau, and ML-driven analytics. Known for improving data reliability, accelerating performance, and driving data-informed decision-making.

Technical Skills

- **Cloud Platforms:** AWS (Kinesis, Lambda, Glue, RDS, Redshift), Azure (Data Factory, Data Lake, Databricks, Synapse)
- **Data Engineering & ETL:** Apache Kafka, Apache Spark, Apache Airflow, Hadoop, AWS Glue, Azure Data Factory
- **Programming & Scripting:** Python (Pandas, PySpark, NumPy), SQL
- **Databases & Storage:** Amazon RDS, AWS Redshift, Azure Synapse, SQL Server, Parquet
- **Data Visualization & BI:** Power BI, Tableau
- **Machine Learning & AI:** Scikit-learn, Metric Learning, AI Chatbots

Professional Experience

Data Engineer, LPL Financial

05/2024 – Present | Remote, USA

- **Designed and deployed a high-throughput, real-time data pipeline using AWS Kinesis**, enabling seamless processing of 500K+ user contacts and transactions, and boosting ingestion performance by 75% for mission-critical systems.
- **Built scalable ETL workflows with AWS Glue and Python (Pandas, PySpark)** to unify disparate datasets, streamline complex transformations, and cut processing time by 60%. Implemented real-time stream processing with AWS Lambda, while utilizing Hadoop for batch processing. Integrated Scikit-learn for anomaly detection, accelerating ML-driven insights and decision-making.
- **Leveraged AWS Lambda for real-time stream processing and Hadoop for batch workloads**, integrating Scikit-learn for anomaly detection to accelerate machine learning insights and proactive decision-making.
- **Developed dynamic Power BI dashboards** for real-time analytics and reporting, increasing operational efficiency by 40% and driving data-informed decisions across stakeholders.

Data Engineer, Tiger Analytics

06/2022 – 07/2023 | Chennai, India

- **Architected a secure, scalable ETL pipeline** to migrate data from SQL Server to Azure Data Lake, boosting regulatory compliance and integration efficiency for critical business operations.
- **Accelerated data ingestion by 60%** using Azure Data Factory with Data Flows and Triggers, while optimizing storage with Parquet format for enhanced performance.
- **Optimized transformation workflows** in Azure Databricks with PySpark and Pandas, cutting processing time by 50% and enabling near real-time analytics.
- **Enhanced scalability and performance** for 500GB+ datasets through Azure Synapse, supporting complex queries and high-speed analytics across business units.
- **Improved data retrieval speed by 50%** via strategic partitioning and indexing in Synapse, significantly elevating enterprise reporting capabilities.
- **Designed a high-performance data warehouse** using denormalization techniques, streamlining BI processes and analytical throughput.
- **Ensured data integrity and reliability** by integrating validation and cleansing logic within Azure Data Factory pipelines.
- **Delivered real-time business insights** through dynamic Tableau dashboards connected to Azure Synapse, empowering stakeholders with actionable KPIs.

Associate Data Engineer, Genpact

05/2021 – 05/2022 | Tirupati, India

- Architected a high-throughput real-time analytics pipeline with Apache Kafka, seamlessly processing over 1 million messages per minute to power immediate, data-driven decisions.
- Accelerated data processing for 500 GB of daily input by optimizing distributed data workflows using Apache Spark, significantly improving pipeline efficiency.
- Built robust ETL systems in Python (Pandas, NumPy) to integrate, cleanse, and transform data from 15+ disparate sources, loading 2 TB of structured data into AWS Redshift for advanced analytics.
- Crafted complex, performance-optimized SQL queries in AWS Redshift using window functions and multi-level joins to analyze over 10 TB of data, driving accurate, high-value reporting.
- Automated 50+ mission-critical data workflows using Apache Airflow, enhancing reliability and reducing operational downtime by 30% through smart scheduling and resilient failure recovery.
- Leveraged AWS Kinesis for scalable real-time stream processing, ingesting 300 MB/sec of event data to support latency-sensitive analytics applications.
- Delivered dynamic Power BI dashboards with live AWS Redshift integration, enabling business stakeholders to access actionable insights in real time and make informed decisions faster.

Education

Master of Science, *University at Buffalo-The State University of New York*

08/2023 - 12/2024 | Buffalo, NY, USA

Data Science

Projects

- AI-Powered Chatbot for Trading Insights Oct 2024**
Built an end-to-end AI chatbot using Flask, LangChain, AstraDB, and Gemini API, integrating a Retrieval-Augmented Generation (RAG) model to deliver real-time trading insights with a 95% response accuracy, handling 500+ concurrent queries.
- Metric Learning for Image Similarity Search Nov 2022**
Developed a self-supervised machine learning pipeline with a hybrid loss function (Triplet Loss and Categorical Cross Entropy) and optimized data pipelines, improving image similarity search precision by 15% and model accuracy by 20%.